

Sri Lanka Institute of Information Technology



SE2030: Software Engineering
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Scrum Roles Assignment

Web-Based Vehicle Rental System – Lanka Ride Rentals

Group ID: 2026-Y2-S1-MLB-B13G2-05

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1. Project Overview

Lanka Ride Rentals is building a web-based vehicle rental system as part of the Software Engineering module. This document covers the Scrum team structure – Product owner, Scrum Master and a six-member Development Team – tied to the personas and core functions from the project proposal. From there, it identifies key user persona and their user stories, organizes them into a prioritized Product Backlog and breaks that backlog into four sprints with detailed task- level time estimates. The goal is to show how Agile / Scrum practices apply across the system’s development lifecycle.

2. Identified User Persona

No	Persona	Description
1	Customer	A renter searches and filters vehicles, view details, makes reservations, pays deposits and fees online and raises enquiries with branch staff.
2	Booking Supervisor	A staff member who approves or denies booking requests, manages vehicle availability and records mileage and fuel at pickup and return.
3	Fleet Coordinator	A staff member who handles vehicle maintenance and service records, marks vehicles under man
4	Finance and Admin Manager	A staff member who monitors invoices, processes refunds, calculates late fees and damage charges and handles financial reporting.
5	Operations Manager	A staff member who oversees multi-branch operations, tacks vehicle distribution and location, and monitors booking status and operational performance.
6	System Administrator	A staff member who manages user accounts, roles, permissions, system configuration and platform scalability.

3. User Stories

I. Customer

- US -01: As a customer, I want to be able to search and filter out the vehicle types, the price, gearbox type, fuel type, branch and data range for finding a suitable vehicle.
- US -02: As a customer, I am interested in viewing all the vehicle details, such as pictures and daily rate before booking to make an informed decision.
- US-03: As a customer, I want to be able to pay booking deposits and rental fees online without compromising my safety, to confirm my booking.
- US-04: As a customer, I want to raise my enquiries with the branch staff so they can examine an address.

II. Booking Supervisor

- US-05: As a Booking Supervisor, I want to be able to see all the information I need to approve a booking on one screen without having to switch back and forth between other screens, as I want to process bookings quickly.
- US-06: As a Booking Supervisor, I want to accept or deny, a booking request, to be able to manage branch vehicle availability.
- US-07: As a Booking Supervisor, I want to be able to write down the mileage and fuel level when I pick up and drop off the vehicle to make sure that there is no unusual fuel consumption, or damage to the vehicle.
- US-08: As a Booking Supervisor, I would like to have the vehicle availability visible in real time, to help me avoid the double booking of vehicles between branches.

III. Fleet Coordinator

- US-09: As a Fleet Coordinator, I want to be able to create, review, update and close maintenance and service records to ensure vehicles remain safe and well maintained.
- US-10: As a Fleet Coordinator, I would like to be notified automatically of time to service or mileage to remind me to service so I don't miss maintenance windows.
- US-11: As a Fleet Coordinator, I want vehicles that are in maintenance to automatically move to the unavailability list and not be booked even by customers.
- US-12: As a Fleet Coordinator, I want to view the complete service history for any vehicle any time as a Fleet Coordinator to inform my fleet management decisions.

IV. Finance and Admin Manager

- US-13: As a Finance Manager I want to see the system generating the invoice for completed transactions so that the billing record would be accurate.
- US-14: As a Finance Manager, I want to have a dashboard with daily, monthly and annual data to easily check revenue and expenditures.
- US-15: As a Finance Manager, I want the system to process refunds for authorized cancelations so that client's payments are processed properly.
- US-16: As a Finance Manager, I want the system to calculate any applicable late fees or damaged chargers when a car is returned so that revenue is protected.

V. Operations Manager

- US-17: As an Operations Manager, I want to generate annual reports on overall performance and branch-based comparisons, so that I can evaluate multi-branch operations.
- US-18: As an Operations Manager, I want to view the availability and real-time location of vehicles in all the three branches of Colombo, Kandy and Galle branches to see the distribution of vehicles.
- US-19: As an Operations Manager, Bookings status – pending, approved, ongoing, completed, cancelled – as an Operations Manager, I want to keep track of the status of my bookings so I can have clarity on the operations of my business.
- US-20: As an Operations Manager, I want to see key operational information on a dashboard rather than having to create reports by hand so that I can make informed decisions.

VI. System Administrator

- US-21: As a System Administrator, I want to restrict features of the system based on the user's role to ensure that staff members have access to only those attributes that are necessary for them to do their job.
- US-22: As a System Administrator, I need to be able to manage accounts and system configurations in a manner that user profiles are securely controlled.
- US-23: As a System Administrator, I want to give users the ability to recover their accounts and reset their passwords to access their accounts.
- US-24: As a System Administrator, I want to easily add new branches, pass a user to the system, so that platform can scale without any significant overhaul.

4. Product Backlog and Priorities

ID	User Story	Persona	Priority
PBI-01	US-21: Implement role-based access control	System Administrator	High
PBI-02	US-22: manage accounts and system configurations	System Administrator	High
PBI-03	US-01: search and filter vehicles by criteria	Customer (Renter)	High
PBI-04	US-08: see real time vehicle availability	Booking Supervisor	High
PBI-05	US-02: Make sure all Vehicle Information is accurate prior to reservation	Customer (Renter)	High
PBI-06	US-05: Display booking approval information on a single screen	Booking Supervisor	High
PBI-07	US-06: Inform the crew that the booking request is accepted or denied, and let them know the reason	Booking Supervisor	High
PBI-08	US-09: Create/update maintenance and service records	Fleet Coordinator	High
PBI-09	US-11: Auto-mark maintenance vehicles an unavailable	Fleet Coordinator	High
PBI-10	US-03: Pay booking deposits and rental fees online	Customer (Renter)	Medium
PBI-11	US-13: Create invoices for finished transactions	Finance Manager	Medium
PBI-12	US-19: Monitor booking status	Operations Manager	Medium
PBI-13	US-18: Real-time location visibility across branches	Operations Manager	Medium
PBI-14	US-07: Record mileage and fuel level upon return	Branch Staff	Medium
PBI-15	US-16: Determine late fees or damage charges	Finance Manager	Medium
PBI-16	US-15: Process refunds for authorized cancellations	Finance Manager	Medium
PBI-17	US-23: Account recovery and password reset	System Administrator	Low
PBI-18	US-10: Automated maintenance reminders	Fleet Coordinator	Low
PBI-19	US-12: Access full-service records anytime	Fleet Coordinator	Low
PBI-20	US-04: Submit support enquiries	Customer (Renter)	Low
PBI-21	US-14: View dashboard with daily/monthly data	Finance Manager	Low
PBI-22	US-20: View operational data on dashboard	Operations Manager	Low
PBI-23	US-17: Generate annual performance reports	Operations Manager	Low
PBI-24	US-24: Add new branches, cars and users	System Administrator	Low

5. Sprint Plan & Task Allocations

The product backlog is divided into 4 sprints. These sprints are created using the framework. Each task is owned by the team that's responsible for the main function listed in the project proposal. The main functions are described in section 6.

Sprint 1: Core System Setup & Vehicle Management

- **Focus:** Authentication, User Profiles, and Fleet Inventory building.
- **Task Allocations:**
 - **System Admin Tasks (PBI-01, PBI-02):** create user login, roles (Admin, Customer, Staff), and database schema. (Assigned to: SAKALASOORIYA S.A.A.A & PAHASARA Y.A.P).
 - **Vehicle Management (PBI-03, PBI-04, PBI-05):** Develop CRUD operations to add records to vehicles, view the vehicles in the fleet, and edit availability. (Owner: SAKALASOORIYA S.A.A.A).

Sprint Backlog – System Admin & Authentication

Sprint ID	User Story	Priority
SPI - 01	US-21: Implement role-based access control	High
SPI - 02	US-22: manage accounts and system configurations	High

Sprint Backlog – User Stories & Priority (Vehicle Management)

Sprint ID	User Story	Priority
SPI – 03	US-01: search and filter vehicles by criteria	High
SPI – 04	US-08: see real time vehicle availability	High
SPI - 05	US-02: Make sure all Vehicle Information is accurate prior to reservation	High

Sprint 01 – Task Breakdown

- **System Admin & Authentication**

Task ID	Task	Est. Hours
T – 01.1	Design Database Schema for Users and Roles	6
T – 01.2	Use safe user registration and login procedures.	8
T – 01.3	Allow role-based access controls (Admin, Customer, Booking Supervisor, Fleet Coordinator, Finance Manager)	6
T – 01.4	Recent developments in the user Management module.	4
T – 01.5	Test Authentication and role-based access.	3

- **Vehicle Management**

Task ID	Task	Est. Hours
T – 02.1	Create the structure for the vehicle inventory database	5
T – 02.2	Develop a CRUD API for vehicle records, including their category, brand, price, deposit and location.	8
T – 02.3	Build Vehicle search & Filter UI (Type, Price, Gearbox, Fuel, Branch, Date Range)	6
T – 02.4	Send out real-time notices of availability of vehicles for use	5
T – 02.5	Test vehicle management module	3

Sprint 2: Booking, Reservations & Maintenance Integration

- **Focus:** The goal is to enable the booking of vehicles and maintain the status of the vehicles.
- **Task Allocations:**
 - **Reservation & Booking (PBI-06, PBI-07, PBI-12):** Create user interface for Reservation & Booking for Customers and user interface for Branch Staff to approve/cancel the booking or change the booking status. (Owner: SAMARANAYAKE P.I.S)
 - **Fleet Maintenance (PBI-08, PBI-09):** Develop logic to maintain and update availability of vehicles in maintenance and prevent double bookings. (Owner: DE SILVA D.L.K.C)

Sprint Backlog – Booking, Reservation & Maintenance Integration

Sprint ID	User Story	Priority
SP2-01	US-05: Display booking approval information on a single screen	High
SPT-02	US-06: Inform the crew that the booking request is accepted or denied, and let them know the reason	High
SPT-03	US-19: Monitor booking status	Medium

Sprint Backlog – User Stories & Priority (Fleet Maintenance Integration)

Sprint ID	User Story	Priority
SP2-03	US-09: Create/update maintenance and service records	High
SP2-04	Auto-mark maintenance vehicles an unavailable	High

Sprint 02 – Task Breakdown

- **Booking & Reservation Maintenance**

Task ID	Task	Est. Hours
T-03.1	T-01.1 UI design for customer bookings (date selection and branch pick-up choice)	6
T-03.2	T-01.2 Development of booking approval screen for branch employees	6
T-03.3	T-01.3 UI design of booking workflow (pending, approved, active, completed, cancelled)	5
T-03.4	T-01.4 Integration of booking module with vehicle availability check system	4
T-03.5	T-01.5 Testing of booking and reservation process	3

- **Fleet Maintenance & Integration**

Task ID	Task	Est. Hours
T-04.1	T-04.1 Design schema for maintenance/ service record	4
T-04.2	T-04.2 Develop CRUD functionality for maintenance/ service records	6
T-04.3	T-04.3 Create auto-unavailable flag for vehicles undergoing maintenance	5
T-04.4	T-04.4 Connect maintenance module to booking module to avoid double booking	4
T-04.5	T-04.5 Test fleet maintenance integration	3

Sprint 3: Payment Billing & Return Processing

- **Focus:** Transactions, Invoices and Vehicle return logic.
- **Task Allocation:**
 - **Payment & Billing (PBI-10, PBI-11, PBI-15, PBI-16):** Secure transaction management, generation of invoices and update of refunds/deposits. (Owner: KAVINDI P.D.N)
 - **Return Processing / Mileage & Fuel Tracking (PBI-14):** Record mileage and fuel level at pickup/return and flag discrepancies. (Owner: SAMARANAYAKE P.I.S)

Sprint Backlog – Payment & Billing

Sprint ID	User Story	Priority
SP3-01	US-03: Pay booking deposits and rental fees online	Medium
SP3-02	US-13: Create invoices for finished transactions	Medium
SP3-03	US-16: Determine late fees or damage charges	Medium
SP3-04	US-15: Process refunds for authorized cancellations	Medium

Sprint Backlog – Return Processing

Sprint ID	User Story	Priority
SP3-05	US-07: Record mileage and fuel level upon return	Medium

Sprint 03– Task Breakdown

- **Payment & Billing**

Task ID	Task	Est. Hours
T – 05.1	Integrate secure payment gateway (sandbox/ test)	8
T – 05.2	Implement deposit and rental fee payment flow	6
T – 05.3	Auto – generate invoices for completed transactions	5
T – 05.4	Implement refund processing for authorized cancellations	5
T – 05.5	Calculate late fees / damage charges upon vehicle return	4
T – 05.6	Test payment and billing module	3

- **Return Processing**

Task ID	Task	Est. Hours
T – 06.1	Design return checklist UI (mileage and fuel level entry)	4
T – 06.2	Implement backend logic to flag unusual fuel / mileage discrepancies	5
T – 06.3	Link return data with damage / late fee calculation	3
T – 06.4	Test return processing flow	2

Sprint 4: Dashboards, Reporting & Customer Support

- **Focus:** Management views, operational insights and customer alerts.
- **Task Allocation:**
 - **Reporting & Admin Dashboard (PBI-13, PBI-21, PBI-22, PBI-23):** Daily annual reporting and display of metrics, on the management dashboard. (Owner: WICKRAMASINGHE R.D.W.K.G.S)
 - **Customer Service & Alerts (PBI-17, PBI-18, PBI-19, PBI-20, PBI-24):** Add email/SMS booking notification functionality and support ticket/enquiry functionality. (Owner: PAHASARA Y.A.P)

Sprint Backlog – Reporting & Admin Dashboard

Sprint ID	User Story	Priority
SP4-01	US-18: Real-time location visibility across branches	Medium
SP4-02	US-14: View dashboard with daily/monthly data	Low
SP4-03	US-20: View operational data on dashboard	Low
SP4-04	US-17: Generate annual performance reports	Low

Sprint Backlog – Customer Support & Alerts

Sprint ID	User Story	Priority
SP4-05	US-23: Account recovery and password reset	Low
SP4-06	US-10: Automated maintenance reminders	Low
SP4-07	US-12: Access full-service records anytime	Low
SP4-08	US-04: Submit support enquiries	Low
SP4-09	US-24: Add new branches, cars and users	Low

Sprint 04 – Task Breakdown

- **Reporting & Admin Dashboard**

Task ID	Task	Est. Hours
T – 07.1	Design dashboard UI (daily / monthly / annual views)	6
T – 07.2	Implement backend report generation logic	8
T – 07.3	Add branch- based comparison and vehicle utilization metrics	5
T – 07.4	Implement real-time location visibility across Colombo, Kandy and Galle branches	5
T – 07.5	Test reporting and dashboard module	3

- **Customer Support & Alerts**

Task ID	Task	Est. Hours
T – 08.1	Implement emails / SMS notification service (bookings, payments, reminders)	6
T – 08.2	Build support ticket submission and tracking UI	5
T – 08.3	Implement account recovery / password reset flow	4
T – 08.4	Add functionality for admins to add new branches and users (Scalability)	4
T – 08.5	Test customer support and notification features	3

6. Task Assignment & ownership per member

a) Product Owner

Member: WICKRAMASINGHE R.D.W.K.G.S

Project Responsibility: Product Vision, Product Backlog prioritization, requirement clarification and acceptance.

b) Scrum Master

Member: SAMARANAYAKE P.I.S

Project Responsibility: Scrum event coordination, progress monitoring and implement management.

c) Development Team Members

- **Developer 01 – Vehicle Management & System Administration**

Member: SAKALASOORIYA S.A.A.A

Project Responsibility: Vehicle Inventory functionality, search & filter UI, real-time availability updates and system administration configuration.

- **Developer 02 – Payment & Billing Management**

Member: KAVINDI P.D.N

Project Responsibility: Payment Processing, invoice generation, late fee / damage charge calculation and refund handling.

- **Developers 3 – Fleet Maintenance & Servicing Management**

Member: DE SILVA D.L.K.C

Project Responsibility: Maintenance and Service record management and booking – module integration to prevent double booking.

- **Developer 4 – UI/UX Design, Customer Service & Quality Assurance**

Member: PAHASARA Y.A.P

Project Responsibility: Customer-Facing Interfaces, Notifications account recovery and system-wide quality assurance testing.